

010 Element Coursework 3000 words (2025 MOD006564 TRI1 F01CAM)

Start Assignment

- Due 5 Dec by 14:00
- Points 100
- Submitting a file upload
- Available until 5 Dec at 14:00

By submitting, I understand that the piece of work submitted will be considered as the final and complete version of my assignment of which I am otherwise the sole author. I understand both the meaning and consequences of plagiarism and that my work has been appropriately attributed unless otherwise stated. I have not knowingly allowed another to copy my work. Please read our [Academic Honesty Policy](https://library.aru.ac.uk/academic/files/Academic%20Honesty%20Policy.pdf)  (<https://library.aru.ac.uk/academic/files/Academic%20Honesty%20Policy.pdf>) here.

Assessment Format:	CW - Report
Mode of Submission:	Canvas: Assignment with Submission Where your assessment requires you to submit work to Canvas or an External Tool, it is not recommended to submit via the Canvas App due to some limitations in the App functionality. We would suggest instead that submission is made on a browser from a computer or mobile device. It is not recommended that you submit your assignment or take a quiz on a cellular or satellite connection. This is because loss of internet connection during assessment is most commonly associated with these services, although we understand that sometimes this is the only option. Wherever possible, we would recommend that assessments are submitted or undertaken on WiFi or hardwired (ethernet) connections. Large files are best submitted on a hardwired internet connection (i.e., not WiFi).
Weighting:	This assessment is worth 100% of the Module mark
Assessment Level:	6 Generic Assessment Criteria and Marking Standards  (https://myaru.sharepoint.com/sites/student-learning-assessment/SitePages/assessment-criteria-and-marking-level-6.aspx)
Element Learning Outcomes:	<i>The element learning outcomes for this assessment are: 1-3</i>

Marking Approach:	Fine-graded
Feedback Release Date:	13/01/2026
Student Support:	If you have other questions that aren't related to your module course content: Ask Aru  _(https://arul.ink/ask-arul)_ is your online tool to ask questions about your life at ARU, use our chatbot, and get in touch with support teams. My ARU  _(http://arul.ink/my-arul)_ is your personalised student intranet and the go-to place for everything you need to know about life at ARU. If you're still unsure, contact our friendly iCentre  _(https://myaru.sharepoint.com/sites/student-help-support/SitePages/icentre.aspx)_ team who can help with many of your questions, or put you in touch with the right team.

You will need to download the following files to complete the assignment:

AI Tech - MOD006564 Coursework 25-26 V 2.0.docx (<https://canvas.anglia.ac.uk/courses/47089/files/6706898?wrap=1>)

A3C Paper.pdf (<https://canvas.anglia.ac.uk/courses/47089/files/6621378?verifier=2vkodViSus5zn7r5bYNW8BFW1zhKxJ8xhe3enicS&wrap=1>)

Assignment Brief – AI Techniques (MOD006564)

Coursework Assessment (Element 010)

Weight: 100% of module mark

Submission: Canvas (ZIP folder containing report & source files)

Module Leader: *Dr. Faraz Janan*

1. Overview

This coursework consists of **three integrated tasks** designed to assess your understanding of core Artificial Intelligence techniques, their implementation, and their application to real-world or rule-based problems.

You will demonstrate both theoretical knowledge and practical modeling skills across AI domains:

- 1. Reinforcement Learning (Task 1)**
- 2. Fuzzy Logic Systems (Task 2)**

The assignment encourages **critical thinking, applied reasoning, and creativity**. It contributes **100%** of the module grade.

2. Learning Outcomes

On successful completion, students will demonstrate:

- **[LO1]** Understanding of rule-based Expert Systems and their fuzzy modifications.
- **[LO2]** Understanding of genetic and reinforcement learning algorithms, including their integration into intelligent systems.
- **[LO3]** Ability to critically analyse problems and apply suitable AI reasoning or fuzzy techniques.

3. Coursework Structure

Task	Title & Description	Weight	Tools / Methods
Task 1	<i>Critical Review of Reinforcement Learning (A3C)</i> – Conduct a scholarly comparison of A3C with other AI methods. Include a literature table of at least 5 peer-reviewed papers highlighting strengths and weaknesses.	50%	Academic research & comparative analysis
	The review will attract a top tier mark if its almost publishable quality with critical insights and reflects of key analytics of the reviewed papers.		
Task 2	<i>Fuzzy Logic Controller</i> – Design and implement a simple fuzzy controller for a truck fuel injection system based on provided parameters. Define membership functions, rules, and demonstrate outputs graphically.	50%	MATLAB / Python with Fuzzy Logic Toolbox / scikit-fuzzy

4. Deliverables

Each student must submit **one ZIP folder** named with their **Student ID** containing:

1. A **report (PDF, 10–12 pages)** covering all three tasks.
2. All **supporting files**, including:
 - Task 2 source code (MATLAB or Python).
 - Any diagrams, figures, or charts.
3. A **short demonstration** of Task 2 and/or Task 3 may be requested.

5. Generative AI Policy

- ~~Task 1: Must be completed without the use of generative AI tools.~~
- ~~Tasks 2: Students may use AI tools for learning, debugging, or experimentation, but not for writing or reporting.~~
- Misuse of AI tools for report writing constitutes **academic misconduct** under ARU regulations.

6. Marking Criteria

Criterion	Description	Weight
Task 1 – Critical Review	Depth, accuracy, and clarity of literature review; quality of comparison and analysis.	50%
Task 2 – Fuzzy Controller	Correct implementation, rule design, and interpretation of results with graphs.	50%
Report Quality	Presentation, structure, clarity, academic style, correct referencing.	Incorporated within each task

7. Guidance on Structure

- Section 1:** Introduction – link tasks to module learning outcomes.
- Section 2:** Task 1 – critical analysis of A3C with literature table.
- Section 3:** Task 2 – fuzzy logic system design, membership functions, rules, and graphs.
- Section 4:** Discussion and conclusions.
- Appendices:** Code listings, figures, tables, or test data.

8. Assessment and Feedback

- You must attend the **Assessment Support Session (Week 9/10)**.
- Feedback will be released on Canvas within **three weeks of submission**.
- Demonstrations may be required during lab sessions.

9. Grading Overview

Band	Descriptor	Indicative Features
70%+ (Distinction)	Excellent	Advanced understanding, clear reasoning, flawless fuzzy design, critical depth.
60–69% (Merit)	Very Good	Sound logic and implementation, good structure, minor weaknesses.
50–59% (Pass)	Adequate	Basic correctness, limited evaluation, partial completion.

Band	Descriptor	Indicative Features
<50% (Fail)	Unsatisfactory	Incomplete, incorrect implementation, lack of clarity or originality.